

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M.Tech. (CE)

December 2013

CE701.01 Operating System Design & Concepts

Date: 30.12.2013, Monday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) How to attach and detach using shared memory segment? Also explain controlling and decontrolling shared memory. [04]
- (b) Explain the process state diagram for multiprogramming environment. [03]
- (c) Does every process contain its own private kernel stack? Why? [02]
- (d) Under what circumstances processes moves from user-mode to kernel-mode. [02]
- Q - 2 (a) What do you mean by process address space? Show the actual mapping of process address space with sample program. [04]
- (b) What resources are used when a thread created? How do they differ from those when a process is created? [04]
- (c) Draw and explain dynamic behavior of a process under the condition when I/O request is made. [04]

OR

- (c) How does the operating system ensure that it retain control over what threads are using the CPU? Explain in detail with an example. [04]
- Q - 3 Answer the following questions. (Any Four) [12]
- (1) Differentiate Interrupts and Exceptions.
 - (2) When `pid=fork()` is called, what are the sequence of operation executes by kernel?
 - (3) What do you mean by IPC? Differentiate named pipe and unnamed pipe.
 - (4) Explain how kernel interacts with processes and devices.
 - (5) What is light-weight processes(LWP)? Differentiate bound and unbound threads.

SECTION – II

Q-4 Answer the following questions.

- (a) Define: Microkernel. What are the advantages of Microkernel over monolithic kernel? [02]
- (b) Write down the scenarios for retrieval of a buffer in buffer cache. [02]
- (c) Write an algorithm for writing to a disk. [02]
- (d) Draw and explain the structure of buffer pool. [02]
- (e) Write and explain an algorithm for open system call. [03]

Q-5 Answer the following questions.

- (a) Explain the advantages & disadvantages of buffer cache. [04]
- (b) Explain algorithms for mount and unmount system call. [08]

OR

- (b) Write & Explain an algorithm for pipe system call in UNIX system. [08]

Q-6 Answer the following questions. (Any Three) [12]

- (a) Explain file system maintenance in detail.
- (b) Explain research scope & motivation for exokernel.
- (c) Write and explain mount system call in detail.
- (d) Compare the permissions a process must have for the following operations and comment.
 - Creating a new file requires write permission in a directory.
 - Creating an existing file requires write permission on the file.
 - Unlinking a file requires write permission in the directory, not on the file.
